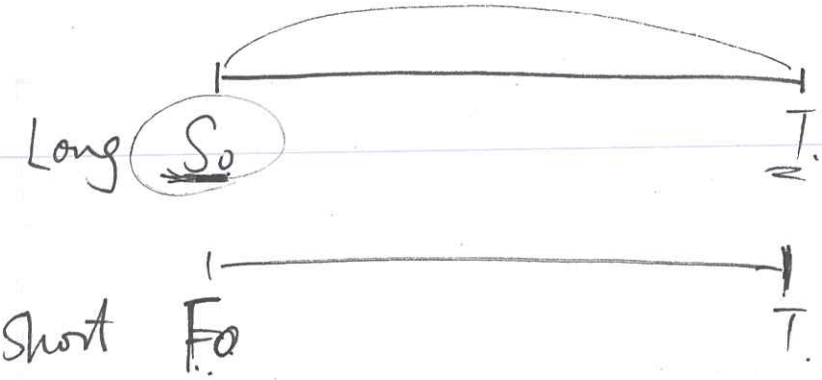
Short Hedge ..



	0	T	
Borrow S_0	$+S_0$	Repay.	$-S_0 \cdot e^{rT}$
Long S_0	$-S_0$	Short S_T	$+S_T$
Short F_0	0	Long F_T	$F_0 - F_T$
Net Position.	0.		$F_0 - S_0 \cdot e^{-rT} = 0$

$$F_0 = S_0 \cdot e^{rT}$$

Cost of Carry.



$$F_0 > S_0 \cdot e^{rT}$$

~~Short = long~~

$F_{(observed)} = \$1100$. $S_0 = \$1000$. $r = 5\%$ $T = 1$.

$$S_0 \cdot e^{rT} = 1000 \cdot e^{0.05 \times 1} = \$1051 < \$1100$$

F_0

1100 − 1051

0		T	
Short F_0	0	Long F_T	~~$F_0 - F_T$~~
Long S_0	$-S_0$	Short S_T	~~$+S_T$~~
Borrowing S_0	$+S_0$	Repayment	$-S_0 \cdot e^{rT}$
	0		$F_0 - S_0 \cdot e^{rT}$

$= 1100 - 1051$

$= \boxed{\$49}$

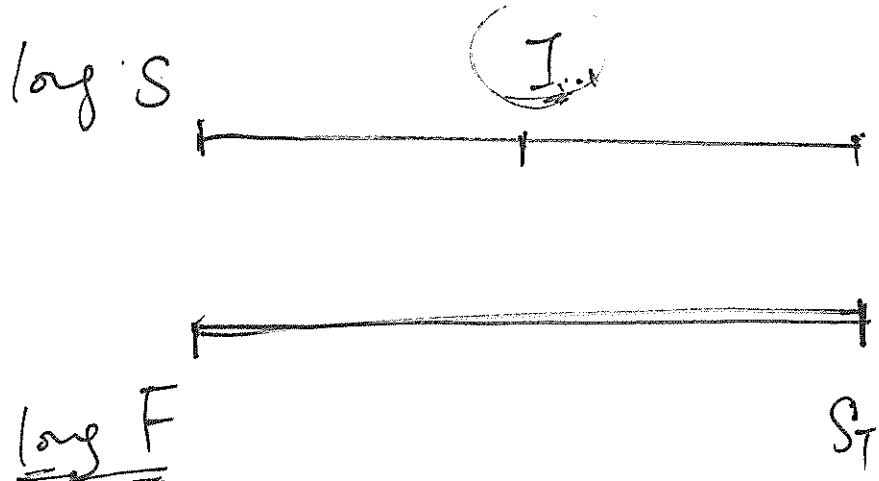
$F_{observed} = \$1000$ $S_0 = \$1000$ $r = 5\%$ $T = 1$.

$F_0 = S_0 \cdot e^{rT} = 1000 \cdot e^{0.05 \times 1} = \$1051$.

$F < S_0 \cdot e^{rT}$.

long short.

0		T	
Long F_0	0.	Short F_T.	~~F_T~~ - 1000
Short Sell S_0.	S_0.	Long S_T	~~$-S_T$~~
Invest in bond. (Cash Investment)	$-S_0$.		$+S_0 \cdot e^{rT}$ $= 1051$.
	0		\$51.



$S_0 \cdot e^{rT}$ $\quad F_0 = (S_0 - I) \cdot e^{rT}$

$$F_0 > S_0 \cdot e^{(r-q)\cdot T}$$

Short　　Long

0		T.	
Short F_0.	0.	Long F_T.	$F_0 - F_T$.
Long S_0.	$-S_0$.	Short S_T	$+S_T$.
Borrow. S_0.	$+S_0$.	Repay. use div. to partly support Repayment	$-S_0 \cdot e^{(r-q)T}$
	0.		$F_0 - S_0 \cdot e^{(r-q)\cdot T} \geq 0$.

$$\underline{F_0} < \underline{S_0 \cdot e^{(r-q)T}}$$

observed theoretical F.

Long Short.

0		T.	
Long F_0	0	Short F_T	$F_T - F_0$
Short Sell S_0.	S_0	Long asset & ret asset	$-S_T$
Invest S_0	$-S_0$	Use investment ret. to pay div. to the lender of the asset.	$S_0 \cdot e^{(r-q)T}$
	0.		

int. income

div returned

$$F_T - F_0 - S_T + S_0 e^{(r-q)T}$$

$$= -F_0 + S_0 \cdot e^{(r-q)T} > 0$$

(3)

(4)

F. D

1 EUR = 1.12 USD

1 BHP = 60 AUD

USD EUR

$F_0 = S_0 \cdot e^{(r - r_f)T}$

$e^{(r-q)}$

U. $T = \frac{2}{12}$

long S_0 $F_0 = (S_0 + U) \cdot e^{rT}$

long F_0 wheat.

$F_0 = S_0 \cdot e^{(r+u-y)T}$

$F_0 < S_0 \cdot e^{(r+u)T}$

Investment.

$$F_0 = S_0 \cdot e^{rT}$$

$$F_0 = (S_0 - I) \cdot e^{rT}$$

$$= S_0 \cdot e^{(r-q)T}$$

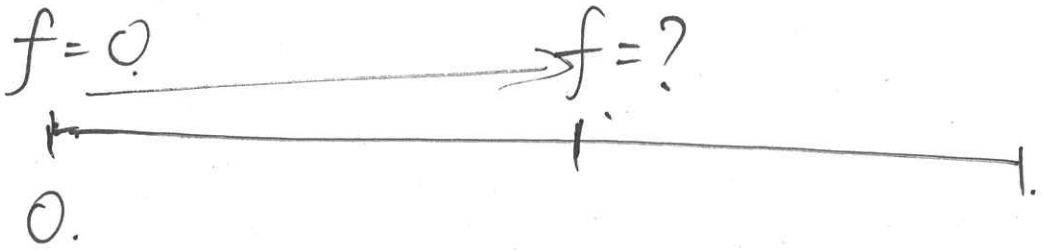
Commodity.

$$F_0 = (S_0 + U) \cdot e^{rT}$$

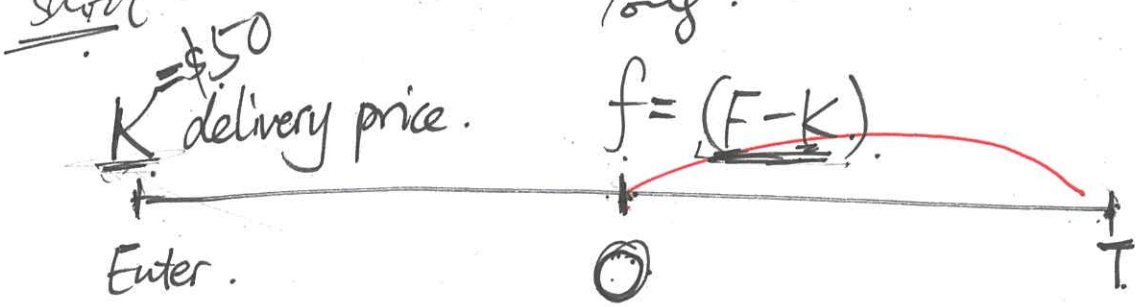
$$= S_0 \cdot e^{(r+u)T}$$

$$F_0 = S_0 \cdot e^{(r+u-y)\cdot T}$$

$E(S_T)$



long
short.



$F = \$55$

~~Forward~~ $f_{(forward)} = (F - K) \cdot e^{-rT}$